Network Issues and Tools

Objectives:

At the end of this episode, I will be able to:

1. Identify common network issues.

Exam Objective: *4.5 - Given a scenario, troubleshoot network connectivity issues.*

Description: In this episode, the viewer will identify the common network issues such as Internet connectivity, unreachable hosts or destinations, resolution issues, DHCP issues as well as the cause of these issues.

- Component Failure
  - Cables (verify cable integrity)
  - NIC (drivers, LED indicators)
  - DNS/DHCP failures (check servers/service)
  - Power issues
- Misconfiguration
  - Misconfigured DHCP options
  - NIC misconfiguration
    - Improper IPv4 configuration
      - IP address (local network communications)
        - ipconfig/ip addr
      - Subnet mask (local network communication)
        - ipconfig/ip addr
      - Gateway address (remote subnet communications, Internet connectivity)
        - ipconfig/ip addr
      - DNS server addresses (Unknown host, FQDN connectivity failure
        - ipconfig/ip addr
  - Improper IPv6 configuration
    - Static vs. Dynamic
  - Firewall misconfiguration
  - VLAN misconfiguration
- Tools and techniques
  - Command
    - ping
      - Destination host unreachable (no route from local system)
      - Ping resource/DHCP/DNS servers
    - tracert/traceroute
      - end to end communication path tracking
    - nslookup/dig
      - Cannot resolve hostname
      - Unknown host
    - nbtstat (older but still used)
    - netstat/nc
      - TCP and UDP connection state information

        - telnet (older but still used)
        - route
            - View correct or incorrect local routes for OS

- Demo
    - DHCP scope issues
        - Show DHCP Scope options configuration
        - Correct DHCP scope options
        - Force DHCP IP address renewal
        - Show updated information
    - IP Address misconfiguration
        - Reminder of DHCP misconfiguration
        - Show TCP/IP properties for NIC
    - NIC failure/non-functional
        - Use Device Manager to locate/diagnosis driver issues
        - Use Device Manager to determine current state NIC
    - Using Route to view local routing table
        - route print

External Resources:

During this episode, you can reference the following external resources for supplementary tools and information:

- Not applicable if blank